

Question number	Answer	Mark
7(a)	A bioinformatics	(1)

Question number	Answer	Mark
7(b)	An explanation that includes the following points: • inhibition of the release of acetylcholine will prevent the generation of a nerve impulse because a chemical is needed to transmit the nerve impulse across the {synapse / gap} (1) • blocking the ion channels will prevent the generation of a nerve impulse because calcium ions will not be able to enter the pre-synaptic bulb and trigger the release of {neurotransmitter / acetylcholine} (1) • blocking the acetylcholine receptors will prevent the generation of a nerve impulse because acetylcholine will not be able to bind to its receptors on the post-synaptic neurone, so {depolarisation of the membrane / initiation of an action potential} will not occur (1)	(3)

Question number	Answer	Mark
7(c)(i)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. Indicative content • neurotoxins have a greater composition of cysteine • non-toxins have a greater composition of glutamine and leucine • the different groups of organisms that produce neurotoxins have similar compositions of leucine • the different groups of organisms that produce neurotoxins have variations in the compositions of cysteine and glutamine • mollusc neurotoxin has the greatest composition of cysteine but bacteria neurotoxin has the least • bacteria neurotoxin has the greatest composition of glutamine and cnidarians the least	(6)

Level	Marks	Descriptor
	0	No awardable content.
1	1-2	Limited number of the most important or relevant scientific factors from the data/information provided are synthesised. No judgement is made.
2	3-4	Some of the most important or relevant scientific factors from the data/information provided are synthesised. A straightforward but accurate judgement is made.
3	5-6	Most of the important or relevant scientific factors from the data/information provided are synthesised. A detailed and accurate judgement is made.

Question number	Answer	Mark
7(c)(ii)	An explanation that includes the following points: • because a different amino acid will have a different R group so different bonds will be formed (1) • therefore the tertiary structure will be different and the neurotoxin will not be able to bind to receptors on the {pre-synaptic membrane /	(2)